

TrueNAS SCALE IoT Dashboard Install Guide

Guided deployment document for installing the IoT Dashboard as a TrueNAS SCALE Custom App, configuring datasets, ports, MQTT, app settings, and safe device discovery.

Important: this guide assumes the dashboard project has been pushed to GitHub without the local .env file. The app is designed for trusted local network use first, not public internet exposure.

1. Pre-install checklist

- " TrueNAS SCALE Apps service is enabled and working.
- " GitHub repository contains Dockerfile, docker-compose.yml, src/, public/, package files, and mosquitto/config/mosquitto.conf.
- " Do not push .env, local SQLite files, data folders, or secrets.
- " Dashboard web port: 3000 TCP.
- " Mosquitto MQTT port: 1883 TCP.
- " IoT subnet: 192.168.30.0/24. IoT gateway: 192.168.30.1.
- " Settings PIN for current test deployment: YOUR_PRIVATE_PIN. Change before broader production use.

2. Create TrueNAS datasets

In TrueNAS SCALE, open Storage > Datasets. Create a parent dataset for app data, then create child datasets for the dashboard database and Mosquitto persistence.

```
apps
apps/iot-dashboard
apps/iot-dashboard/data
apps/iot-dashboard/mosquitto-config
apps/iot-dashboard/mosquitto-data
apps/iot-dashboard/mosquitto-log
```

Example full paths, replacing YOUR_POOL with your actual pool name:

```
/mnt/YOUR_POOL/apps/iot-dashboard/data
/mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-config
/mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-data
/mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-log
```

These datasets keep the SQLite database and MQTT broker data persistent across container upgrades and restarts.

3. Add Mosquitto configuration

Create this file on TrueNAS:

```
/mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-config/mosquitto.conf
```

Use this content for the first trusted local deployment:

```
listener 1883 0.0.0.0
allow_anonymous true
persistence true
persistence_location /mosquitto/data/
log_dest stdout
```

Security note: anonymous MQTT is acceptable only for a trusted local/test network. Later, add MQTT users/passwords and restrict port 1883 to trusted LAN clients.

4. Configure the TrueNAS Custom App YAML

In TrueNAS SCALE, go to Apps > Discover Apps > Custom App and choose the custom YAML / Docker Compose style deployment. Replace YOUR_GITHUB_USERNAME, YOUR_REPO_NAME, YOUR_POOL, and TRUENAS_IP before saving.

```
services:
  iot-dashboard:
    build:
      context: https://github.com/YOUR_GITHUB_USERNAME/YOUR_REPO_NAME.git
      container_name: iot-dashboard
    environment:
      DATABASE_URL: file:/data/iot-dashboard.sqlite
      SETTINGS_PIN: "YOUR_PRIVATE_PIN"
      IOT_ACCESS_MODE: routed
      IOT_SUBNET_CIDR: 192.168.30.0/24
      IOT_SUBNET_NOTES: "TrueNAS/app may initiate restricted connections to IoT devices;
      IoT devices should not initiate LAN access."
      MQTT_HOST: mqtt
      MQTT_PORT: 1883
      MQTT_USERNAME: ""
      MQTT_PASSWORD: ""
      MQTT_USE_TLS: "false"
      NEXT_PUBLIC_APP_URL: http://TRUENAS_IP:3000
    ports:
      - "3000:3000"
    volumes:
      - /mnt/YOUR_POOL/apps/iot-dashboard/data:/data
    restart: unless-stopped
    depends_on:
      mqtt:
        condition: service_healthy
    healthcheck:
      test: ["CMD-SHELL", "wget -qO- http://127.0.0.1:3000/api/health >/dev/null 2>&1 ||
      exit 1"]
      interval: 30s
      timeout: 5s
      retries: 5
      start_period: 20s
  mqtt:
    image: eclipse-mosquitto:2
    container_name: iot-dashboard-mqtt
    ports:
      - "1883:1883"
    volumes:
      - /mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-config:/mosquitto/config
      - /mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-data:/mosquitto/data
      - /mnt/YOUR_POOL/apps/iot-dashboard/mosquitto-log:/mosquitto/log
    restart: unless-stopped
    healthcheck:
      test: ["CMD-SHELL", "nc -z 127.0.0.1 1883"]
      interval: 30s
      timeout: 5s
      retries: 5
      start_period: 5s
```

5. Ports and network access

- " IoT Dashboard: 3000/TCP, exposed only to trusted LAN clients.
- " Mosquitto MQTT: 1883/TCP, exposed only to trusted LAN/app/MQTT devices.
- " Dashboard URL: http://TRUENAS_IP:3000.
- " Allow TrueNAS/App to initiate traffic to 192.168.30.0/24.
- " Block IoT subnet initiated access to the main LAN except established replies.

```
Allow: TrueNAS/App -> 192.168.30.0/24
Block: IoT subnet -> main LAN, except established replies
```

6. Start and verify the app

After saving the Custom App YAML, deploy/start the app from TrueNAS. Watch the logs for both containers: `iot-dashboard` and `iot-dashboard-mqtt`.

On first start, the dashboard automatically creates the SQLite database and schema here:

```
/mnt/YOUR_POOL/apps/iot-dashboard/data/iot-dashboard.sqlite
```

Open a browser from a trusted client device:

```
http://TRUENAS_IP:3000
```

Useful pages:

```
http://TRUENAS_IP:3000
http://TRUENAS_IP:3000/devices
http://TRUENAS_IP:3000/settings
http://TRUENAS_IP:3000/display
```

7. Unlock and configure Settings

Open Settings and enter the PIN:

```
http://TRUENAS_IP:3000/settings
PIN: YOUR_PRIVATE_PIN
```

- " Network & IoT access: Access mode routed; IoT subnet 192.168.30.0/24; gateway note 192.168.30.1.
- " MQTT broker: Host mqtt; Port 1883; TLS false; username/password blank for current local Mosquitto config.
- " Discovery: Start with mock mode. Switch to live only after dashboard and network access are confirmed.
- " Control safety: Keep risky controls disabled until each integration is confirmed safe.

8. Run diagnostics

From Settings, run the available diagnostics in this order:

1. Test MQTT
2. Check IoT Gateway
3. Sample MQTT Topic

Expected results: MQTT should connect when the Mosquitto container is running. Gateway check should succeed when TrueNAS can reach 192.168.30.1. MQTT sample may return no message unless a device is actively publishing to the configured sample topic.

9. Scan for devices safely

Open Devices > Discovery. For the first pass, keep Discovery Mode set to mock and run discovery to confirm the UI workflow. Then return to Settings, switch Discovery Mode to live, and run discovery again.

The live scanner is intentionally cautious. It scans a limited range and probes common ports only, including 80, 443, 1883, 502, 8081, and 6668. This avoids aggressive scanning on a test deployment.

After discovery, review each device, confirm the likely type, then approve it into the dashboard. Do not mark a device as controllable unless the integration and risk level are understood.

10. Add known Wi-Fi devices manually

Some Wi-Fi devices will not identify reliably through network scanning. For those, use **Devices > Add known Wi-Fi device**. Enter the name, IP address, room, category, integration, risk level, and any MQTT topics or notes. DHCP reservations are recommended for important devices.

11. Recommended production hardening

- " Change SETTINGS_PIN from YOUR_PRIVATE_PIN before wider use.
- " Add Mosquitto username/password and update app MQTT settings.
- " Expose ports 3000 and 1883 only to trusted local networks.
- " Back up /mnt/YOUR_POOL/apps/iot-dashboard/data regularly.
- " Do not expose directly to the internet without proper reverse proxy, TLS, authentication, and firewall rules.

12. Troubleshooting quick checks

- " Dashboard does not load: confirm app container is running and port 3000 is not already used.
- " Settings or Devices errors: confirm /data volume is writable and SQLite file was created.
- " MQTT test fails: confirm mqtt container is healthy and mosquitto.conf exists in the mounted config dataset.
- " Gateway check fails: confirm TrueNAS routing/firewall allows traffic to 192.168.30.1.
- " No devices found: start with manual known devices, confirm live discovery mode, and verify IoT subnet routing.

End state

When complete, the IoT Dashboard should run as a persistent TrueNAS SCALE Custom App with SQLite storage under the app dataset, a local Mosquitto broker, PIN-protected Settings, responsive web access from LAN clients, and safe guided onboarding for known and discovered IoT devices.